What has to be proved

A tiered catalogue of the statements behind the irfsolve classification

Supervisor notes for the rewrite

6 October 2026

Abstract

This note collects every mathematical statement on which the irfsolve pipeline silently relies, as documented in the implementation reference, and sorts them into five tiers by how much proof they need: definitions and bookkeeping (A), elementary lemmas (B), structural theorems (C), imported results (D), and steps that are not theorems at all (E). Each item names the milestone of the development plan at which the code first depends on it and the functions of the reference that use it, and flags the step where an error is most likely. Three findings go beyond the reference: the identity $Q_5 = -[B_H, H]$, which places the condition $[H, Q_5] = 0$ under Hokkyo's theorem; the correct chain length for a periodic re-check, which is $L \geq 9$ and not $L = 7$; and the precise sense in which the exact triviality tests do or do not speak about real points.

1 How to read this

Tiers. **A**: one-line proofs, but the code depends on the convention being fixed. **B**: a few lines; errors here are signs, index ranges, off-by-one. **C**: real proofs; the soundness of the pipeline rests on them. **D**: standard results to be cited, not reproved. **E**: probabilistic, numerical or empirical steps that must be labelled as such in the output.

Tags. M1–M5 refer to the milestones of the development plan. Items marked OUT OF SCOPE only support features that the plan (§5) excludes: symmetry orbits, local-charge counting, `facstd`, the `'irf'` correction. They can be skipped unless the feature is added.

Scope of the supervisor's contribution. In line with the development plan, the items give statements and the critical step, not proofs or implementation recipes. Short derivations are included only where the statement is new relative to the reference (B3, B7, C5). The two questions that the plan reserves for the student (what counts as trivial, and when two solutions are the same model) appear only as open obligations in Section 7.

Notation. $M_d = \mathbb{C}\mathbf{1} \oplus M_d^0$ (identity plus traceless part). For a density D on n sites, $\mathcal{T}(D) := \sum_j T^j D$ is the formal translation-invariant (TI) sum. $\tau(G) := G \otimes \mathbf{1} - \mathbf{1} \otimes G$ is the telescoping density of G. $\mathrm{wrap}_L(D)$ is the same sum on a ring of L sites, with D placed on sites $0, \ldots, n-1$. $r(\cdot)$ denotes range. $h = \sum_{a,b} P^a \otimes h^{ab} \otimes P^b$ is the IRF density, $H = \mathcal{T}(h)$, and $Q_5 = \sum_j [h_j, h_{j+1}] = \mathcal{T}\big([h \otimes \mathbf{1}, \mathbf{1} \otimes h]\big)$. $G_\gamma = \sum_j g_{j,j+1}$ with $g = \sum_{a,c} \gamma_{ac} P^a \otimes P^c$ is the `'diag-gauge'` correction.

2 Tier A — definitions and bookkeeping

A1 (definition of a TI operator; M2)**.** Let $\{e_0 = \mathbf{1}, e_1, \ldots, e_{d^2-1}\}$ be a basis of M_d with $e_\alpha \in M_d^0$ for $\alpha \geq 1$, and expand $D = \sum_u D_u \, e_{u_1} \otimes \cdots \otimes e_{u_n}$. A *core* is a word w whose first and last letters

are non-zero (interior zeros allowed); strip(u) deletes the leading and trailing zeros of u. Put

$$\kappa_w(D) = \sum_{u:\ \mathrm{strip}(u)=w} D_u, \qquad \kappa_\emptyset(D) = D_{0\cdots 0}.$$

Definition: $\mathcal{T}(D) = \mathcal{T}(D')$ iff $\kappa_w(D) = \kappa_w(D')$ for every core w and for $w = \emptyset$. Hokkyo [1, §III] gives the rigorous model: a TI operator assigns to each finite $A \subset \mathbb{Z}$ a component in $\bigotimes_{i\in A} M_d^0$, and commutators are defined locally; κ, grouped by the set of non-identity positions, is exactly that component.

Critical step. Every statement of the form "vanishes on the infinite chain" is relative to this definition. With it, uniqueness in C2 is almost tautological; the content moves to C1 and to the correctness of the recursion.

A2 (padding; M2)**.** $\mathcal{T}(X \otimes \mathbf{1}) = \mathcal{T}(X) = \mathcal{T}(\mathbf{1} \otimes X)$; consequently $\mathcal{T}(\tau(G)) = 0$.

Used in: `pad`, right-padding inside `ti_comm`.

A3 (matrix units; M2)**.** For multi-indices, $E_{rc}E_{r'c'} = \delta_{cr'}E_{rc'}$, and $\mathbf{1} = \sum_c E_{cc}$.

Used in: `mul`, `pad`.

A4 (exact affine data; M1/M3)**.** Every entry of an ansatz block is $c_0 + \sum_k c_k x_k$ with $c_k \in \mathbb{Q}(i)$. Hence the real and imaginary parts of every density entry lie in $\mathbb{Q}[x]$ and have degree at most one. Replacing a symbol x by $\sqrt{2}\,x$ (spin-1 normalisation) is a linear change of variables and maps solution sets bijectively.

Used in: `IRFAnsatz.__init__`, `_check_rational`, `EquationSystem._entry`.

A5 (real split; M3)**.** For $x \in \mathbb{R}^n$ and $P, Q \in \mathbb{Q}[x]$: $P(x) + iQ(x) = 0 \iff P(x) = Q(x) = 0$.

Used in: `generate`.

A6 (ideal-preserving operations; M3)**.** Multiplying generators by non-zero rationals (monic normalisation), deleting duplicates, and replacing a list by an rref basis of its $\mathbb{Q}$-span leave the ideal unchanged. Replacing linear forms by an rref basis of their span leaves their common zero set unchanged.

Used in: `generate`, `linear_reduce`, `_independent_linear`.

A7 (restriction; M1/M4)**.** Substituting a parametrisation of an affine subspace A gives a bijection between the restricted solution set and $V \cap A$. Building an activity pattern into the ansatz is equivalent to adding the inactive blocks' forms as equations.

Used in: `restrict`, `ab_hermitian`.

A8 (activity; M4)**.** All forms of block (a, b) vanish iff the block is diagonal. $V = \bigsqcup_S V_S$, because every point has exactly one set of active blocks. Blocks with identical form spaces are active simultaneously, so a pattern separating them is empty.

Used in: `_forms`, `_consistent`, `stratum`.

A9 (classical stratum; M4)**.** If every block is diagonal, then $Q_5 = 0$ and $[H, G_\gamma] = 0$; the whole diagonal locus solves the condition, with or without correction.

Used in: `patterns` (skipping $S = \emptyset$).

A10 (saturation order; M4)**.** $I \subseteq I : J^\infty$, hence $1 \in I \Rightarrow 1 \in I : J^\infty$.

Used in: `probe_chart`.

3 Tier B — elementary lemmas

B1 (TI commutator; M2)**.** For A on a sites and B on b sites,

$$[\mathcal{T}(A), \mathcal{T}(B)] = \mathcal{T}(C), \qquad C = \sum_{s=-(b-1)}^{a-1} [A_0, B_s],$$

where each term lives on a window of length at most $a+b-1$; padding all terms to any common $N \geq a+b-1$ leaves $\mathcal{T}(C)$ unchanged (A2). For the residual, $(a, b) = (3, 4)$ and $N = 6$.

Critical step. The shift range is inclusive at both ends. Dropping a single shift silently removes equations, and the weaker system then has spurious solutions.

Used in: `ti_comm`, `generate`.

B2 (commutators have no identity part; M2)**.** $\mathrm{tr}[X, Y] = 0$, hence $\kappa_\emptyset$ of any commutator density vanishes. This is the only justification for discarding the scalar at $\ell = 1$ in `canonical` (defect D15 of the reference): on general input the identity component would be lost.

B3 (IRF reduction of the charge; M3)**.** For IRF densities $[h_j, h_{j+2}] = 0$, because the only shared site is a control site of both factors, on which both act diagonally. With the boost $B_H = \sum_j j\, h_j$, grouping the unordered pairs $\{j, j+s\}$, $s \geq 1$, gives

$$[B_H, H] = \sum_{j,k} j\,[h_j, h_k] = \sum_j \sum_{s\geq 1} \big(j - (j+s)\big)[h_j, h_{j+s}] = -\sum_j \Big([h_j, h_{j+1}] + 2\,[h_j, h_{j+2}]\Big) = -Q_5.$$

Hence the `'IRF'` and `'GP'` forms of the charge coincide for this ansatz, and $[H, Q_5] = 0$ is exactly the boost condition of Hokkyo's theorem (D9), with $Q_5 = -Q^{(3)}$ in his notation.

Critical step. For a general three-site density the coefficient of $[h_j, h_{j+2}]$ is 2 in the boost charge and 1 in the Gombor–Pozsgay form [3]; the two agree here only because $[h_j, h_{j+2}] = 0$. (Checked numerically, Appendix A.)

B4 (Hermiticity and the partner skip; M3)**.** At real x: $H^\dagger = H$, $Q_5^\dagger = -Q_5$, hence $[H, Q_5]^\dagger = [H, Q_5]$; likewise $[H, [H, G]]$ is Hermitian for Hermitian G. The canonical map commutes with $\dagger$ (partial trace and $\mathbf{1} \otimes \cdot$ do, and $1/d$ is real), so every c_ℓ is Hermitian. At the polynomial level,

$$\mathrm{Re}\, c_\ell[c, r] = \mathrm{Re}\, c_\ell[r, c], \qquad \mathrm{Im}\, c_\ell[c, r] = -\,\mathrm{Im}\, c_\ell[r, c] \quad \text{in } \mathbb{Q}[x],$$

because both sides agree on $\mathbb{R}^n$. Skipping transposed partners therefore preserves the ideal exactly, including at complex points.

Critical step. This must be re-proved for every term added to the residual. Defect D1 (`'irf'` with a Hermitian $\tilde{h}$, giving an anti-Hermitian $[H, \tilde{H}]$) is precisely this failure.

Used in: `generate` (extraction loop).

B5 (homogeneity and cones; M3/M4)**.** If no ansatz entry has a constant term, the residual is a homogeneous cubic in h. With `'diag-gauge'` it is homogeneous of degree 3 under the joint scaling $(h, \gamma) \mapsto (\lambda h, \lambda\gamma)$. Block and two-local forms are linear, vacuity forms quadratic, all homogeneous. Hence V and every V_S are stable under $\lambda \in K^\times$, $K \in \{\mathbb{R}, \mathbb{C}\}$.

Critical step. This fails as soon as an entry has a constant term (defect D11), and γ must carry weight one.

B6 (charts; M4)**.** Let I be homogeneous in the sense of B5 and let $F_1, \ldots, F_m$ be the linear forms of the first active block. Then

$$\bigsqcup_{c=1}^{m} V\big(I + \langle F_1, \ldots, F_{c-1}, F_c - 1\rangle\big) \;\longrightarrow\; \big(V(I) \setminus V(F_1, \ldots, F_m)\big)/K^\times$$

given by $x \mapsto [x]$ is a bijection for $K = \mathbb{R}$ and $K = \mathbb{C}$, and real points correspond to real points. The dimension of a cone component equals the dimension of its chart plus one.

Used in: `stratum`.

B7 (telescoping gauge; M3)**.** For a diagonal two-site g, the density $h' = h + \tau(g)$ is again IRF, $H' = H$, and

$$Q_5(h') = Q_5(h) + [H, G], \qquad G = \sum_j g_{j,j+1}, \qquad [H, G] = \mathcal{T}\big([h,\ g \otimes \mathbf{1} + \mathbf{1} \otimes g]\big).$$

Conversely, $\tau(g)$ has IRF shape only if $g \in \mathrm{Diag} \otimes \mathrm{Diag}$: off-diagonal content of the first factor of g appears on a control site of $\tau(g)$ and cannot cancel.

Critical step. $[h_k, g_j] \neq 0$ only for $j \in \{k, k+1\}$; this uses that the outer sites of h_k and both sites of g are diagonal. (Checked numerically for $d = 2, 3$, Appendix A.)

Used in: `aux_density`, `gauge_charge_extra`.

B8 (meaning of the correction; M3)**.**

$$\exists \gamma :\ \big[H,\ Q_5(h) + [H, G_\gamma]\big] = 0 \quad \Longleftrightarrow \quad \exists\, g \in \mathrm{Diag}^{\otimes 2} :\ \big[H,\ Q_5(h + \tau(g))\big] = 0.$$

Moreover $Q_5 + [H, G] = -[B_H + G,\ H]$, and $B_H + G$ is again a boost ($T(B_H + G) - (B_H + G) = H$ because G is TI), so D9 covers the corrected condition as well. The solution set is invariant under $(h, \gamma) \mapsto (h + \tau(g), \gamma - g)$ and under $h \mapsto h + c\mathbf{1}$ (for every correction option). Components are therefore cylinders along these directions, which enters every dimension count.

B9 (linearity of the normal form; M2/M3)**.** The recursion of `canonical` uses only addition, subtraction and multiplication by $1/d$. It is therefore $\mathbb{Q}[x]$-linear and commutes with evaluation at any point. This is what turns "the entries of c_ℓ" into "equations whose common zeros are exactly the points where the residual vanishes".

B10 (symmetry transport; OUT OF SCOPE)**.** Reflections and permutation relabelings map IRF solutions to IRF solutions, V_S to $V_{\sigma S}$, and the diag-gauge space to itself. Only permutation matrices preserve diagonality, and hence activity. The family itself must be invariant (`check_symmetries` tests this only probabilistically).

B11 (commutant; OUT OF SCOPE)**.** For Hermitian H, $[H, Q] = 0 \Rightarrow [H, Q^\dagger] = 0$; hence the complex dimension of the commutant equals the real dimension of its Hermitian part.

4 Tier C — structural theorems

C1 (telescoping kernel; M2)**.** For densities on n sites, $\mathcal{T}(D) = 0 \iff D = \tau(G)$ for some G on $n - 1$ sites. The direction $\Leftarrow$ is A2; $\Rightarrow$ is the content, and it is the precise form of "removing telescopically vanishing components". Proved in the canonical-form note.

C2 (normal form; M2)**.** Every TI operator of range at most n has a unique representation

$$\lambda\, \mathcal{T}(\mathbf{1}) + \sum_{\ell=1}^{n} \mathcal{T}(c_\ell), \qquad c_\ell \in M_d^0 \otimes M_d^{\otimes(\ell-2)} \otimes M_d^0 \quad (c_1 \in M_d^0),$$

with $\lambda = \mathrm{tr}\, D / d^n$; it vanishes iff $\lambda = 0$ and all $c_\ell = 0$, and the recursion of `canonical` computes exactly these c_ℓ, independently of the padding length.

Critical step. The output must have traceless ends on *both* sides; $\mathrm{tr}_1 d_2 = 0$ needs $\mathrm{tr}_1 d_1 = 0$. The bookkeeping $\mathcal{T}(D) = \mathcal{T}(d_2) + \mathcal{T}\big((r+s)/d\big)$ must be exact, including the factor $1/d$ at every level.

Corollaries (one line each once C2 holds):

(i) two-local test: $c_3(h) = 0 \iff r(H) \le 2$;
(ii) vacuity test: $c_\ell\big([h \otimes \mathbf{1}, \mathbf{1} \otimes h]\big) = 0$ for all $\ell \iff Q_5 \equiv 0$;
(iii) numerical canonical coordinates vanish iff the operator does, with placement-independent scale $\sqrt{d}^{\,N-\ell}$;
(iv) OUT OF SCOPE: strings of length r with non-identity first letter form a basis of TI operators of range at most r, modulo $\mathbf{1}$.

C3 (equation correctness; M3 master theorem)**.** Let E be the generated list. Then

$$\langle E \rangle = \big\langle \text{Re, Im of all entries of all } c_\ell(R) \big\rangle \subset \mathbb{Q}[x],$$

and for $x \in \mathbb{R}^n$ the (corrected) residual vanishes on the infinite chain iff $E(x) = 0$. Proof by assembly of B1, B2, B4, B9, C2, A5, A6. The ideal-level statement is what the computer algebra system needs; the pointwise statement is the physics.

C4 (search completeness; M4 master theorem)**.** Every real solution outside the classical stratum appears, up to $\mathbb{R}^\times$, in exactly one chart of exactly one stratum, and lies on some $P \in \mathrm{Min}(I_{\text{chart}})$ with $J \not\subseteq P$, i.e. on a component that is not excluded. (If every minimal prime through x contained J, then $x \in V(J)$, contradicting activity.) Proof by assembly of A8, B5, B6 and D3–D5.

Critical step. Each step is easy, but this is the claim "nothing was lost" that the classification makes; it should be written down. If the modular probe or `modStd(I,0)` is used, the statement holds only up to E1–E2.

C5 (periodic comparison; M1/M5)**.** Let Φ, Ψ be TI with densities of ranges a, b, and $n := a+b-1$.

(a) Forward direction. For $L \ge n$, $[\mathrm{wrap}_L \Phi, \mathrm{wrap}_L \Psi] = \mathrm{wrap}_L[\Phi, \Psi]$: the union of the supports of every overlapping pair has length at most $n \le L$, so no term meets itself around the ring. Hence vanishing on the infinite chain implies vanishing on every ring with $L \ge n$ (here $L \ge 6$). The acceptance test of M1 uses only this direction and is sound as designed.

(b) General densities. On TI operators of range at most n, wrap_L is injective iff $L \ge 2n-1$. At $L = 2n-2$ and for linearly independent $X, Y \in M_d^0$,

$$\mathcal{T}\big(X \otimes \mathbf{1}^{\otimes(n-2)} \otimes Y - Y \otimes \mathbf{1}^{\otimes(n-2)} \otimes X\big) \neq 0,$$

but its two terms have the same image on the ring,

$$\sum_{j \in \mathbb{Z}_L} Y_j X_{j+n-1} = \sum_{k \in \mathbb{Z}_L} X_k Y_{k-(n-1)} = \sum_{k \in \mathbb{Z}_L} X_k Y_{k+(n-1)},$$

since $-(n-1) \equiv n-1 \pmod{2n-2}$.

(c) Commutators. A core of maximal length n in $[\Phi, \Psi]$ comes only from the two extremal shifts, where the supports overlap in a single site; the factor there is a commutator of single-site operators, hence traceless. So every length-n core of $[\Phi, \Psi]$ has a non-identity letter at position $a-1$ or $b-1$. Hokkyo [1, Cor. 19] proves injectivity on commutators whenever $a + b \le L/2 + 2$, which for $(a, b) = (3, 4)$ gives $L \ge 10$. The counting argument below sharpens this to $L \ge 9$ for $(3, 4)$, independently of d: the kernel of wrap_L is spanned by differences of cores with the same cyclic word; at $L = 9$ a cyclic word has two readings of length ≤ 6 only if it has two identity runs of length ≥ 3, and enumerating these words shows that every such orbit contains exactly two cores, at least one of which has length 6 with identities at positions 2 and 3. No $(3, 4)$ commutator contains such a core, so the kernel meets the commutator span trivially. For $d = 2$ the bound is sharp: on the span of all $(3, 4)$ commutators, wrap_9 is injective while wrap_8 has a 50-dimensional kernel (Appendix A).

Consequence. A periodic check of $[H, Q_5] = 0$ (with or without `'diag-gauge'`) certifies the infinite-chain identity only for $L \ge 9$; at $L = 7$ or 8 it can refute the identity but not certify it. For qutrits $3^9 = 19\,683$ basis states, well within reach of sparse matrices.

Critical step. The reference's sentence in `verify_periodic` ("$L = 7$ exceeds the range 6 of $[H, Q_5]$, so no wrap-around cancellation can hide a violation") conflates (a) with (b)–(c) and is false.

C6 (corrected vacuity is a rank condition; M5, `'diag-gauge'` only)**.** Fix h; let $t(h)$ be the canonical coordinates of Q_5 and let $M(h)$ have as columns those of $[H, G_{ac}]$. Then $\{\gamma : Q_5 + [H, G_\gamma] \equiv 0\}$ is an affine space, non-empty iff

$$\operatorname{rank} M(h) = \operatorname{rank} \big[M(h) \,\big|\, t(h) \big].$$

On an irreducible component both ranks take their generic values on a dense Zariski-open set, and only there does a pointwise test decide the component.

Critical step. The statement is existential in γ; it is not "the residual vanishes at the component's own γ". The exact vacuity test of `solve_chart` does not cover it (defect D10).

Used in: `vacuous_up_to_correction`.

C7 (real points versus complex components; M5)**.** Let $P \subset \mathbb{Q}[x]$ be prime and $p \in V_{\mathbb{R}}(P)$ with $\operatorname{rank} J_P(p) = n - \dim P$, where J_P is the Jacobian of a generating set of P itself. Then the geometric component $W \ni p$ is unique and conjugation-stable, $W(\mathbb{R})$ is Zariski-dense in W, and for $T \subset \mathbb{Q}[x]$:

$$T \text{ vanishes on } W(\mathbb{R}) \text{ near } p \iff T \subseteq P.$$

Without such a point, $V_{\mathbb{R}}(P)$ may be empty or lie in a proper subvariety (for example in the intersection of conjugate components), and the exact class of the component can differ from the class of every real point. This is the phenomenon observed in family D.

Critical step. Use the generators of P returned by `minAssGTZ`, not those of the original system; the corank computed from the original equations (`local_dim`) is only an upper bound on the dimension.

Used in: `analyse_component`.

5 Tier D — imported results

D1 (membership; M4)**.** For a Gröbner basis G of I, $f \in I \iff \mathrm{NF}_G(f) = 0$ [4]. Used by the `reduce` tests.

D2 (Nullstellensatz over $\mathbb{Q} \subset \mathbb{C}$; M4)**.** $V_{\mathbb{C}}(I) = \emptyset \iff 1 \in I$, and $I_{\mathbb{Q}}(V_{\mathbb{C}}(I)) = \sqrt{I}$ [4]. Hence "dim $= -1$" means no complex, and therefore no real, solutions; for prime P, $T \subseteq P$ iff T vanishes on $V_{\mathbb{C}}(P)$; for a non-radical piece, membership is sufficient but not necessary for vanishing.

D3 (saturation; M4)**.** $V(I : J^\infty) = \overline{V(I) \setminus V(J)}$; $(I : J_1^\infty) : J_2^\infty = I : (J_1 J_2)^\infty$ and $V(J_1 J_2) = V(J_1) \cup V(J_2)$ [4].

D4 (minimal primes of a saturation; optional)**.** $\mathrm{Min}(I : J^\infty) = \{P \in \mathrm{Min}(I) : J \not\subseteq P\}$ [5]. Needed only for saturation-by-test, which the plan (§6) lists as optional.

D5 (decomposition; M4)**.** `minAssGTZ` returns $\mathrm{Min}(I)$, and $V(I) = \bigcup_{P \in \mathrm{Min}(I)} V(P)$ [6, 5]. (`facstd` pieces are not necessarily prime: OUT OF SCOPE.)

D6 (Galois transitivity; M5)**.** For a prime $P \subset \mathbb{Q}[x]$, $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ permutes the geometric components of $V(P)$ transitively; a rational polynomial vanishes on one of them iff it vanishes on all. Used in C7.

D7 (lucky primes; M4)**.** For fixed $I \subset \mathbb{Q}[x]$, all but finitely many primes p have the property that the reduced Gröbner basis of I reduces modulo p to that of I_p [7, 8]. In particular emptiness is detected correctly modulo almost every prime, but the bad primes are not known in advance.

D8 (real points; M5)**.** An irreducible real algebraic variety has Zariski-dense real points iff it has a non-singular real point [9]. Used in C7. If "real points exist" is to be a theorem rather than evidence, an approximate zero must be certified (Smale's α-theory or Newton–Kantorovich [10]).

D9 (Reshetikhin condition implies a commuting hierarchy; M5)**.** [1, Thm. 6] (preprint). For a Hermitian, TI, finite-range H and any finite-range boost B (i.e. $T(B) - B = H$),

$$[H,[B,H]] = 0 \quad \Longrightarrow \quad (\mathrm{ad}_B)^{n-1} H \text{ is TI and conserved for every } n \geq 1,$$

and these operators mutually commute. Through B3 and B8 this applies to both `'none'` and `'diag-gauge'`; it upgrades the Grabowski–Mathieu conjecture [2] to a theorem in this sense.

Critical step. The theorem gives a commuting hierarchy, not its non-triviality. The paper's statements that the hierarchy has growing range rest on non-degeneracy assumptions formulated for nearest-neighbour interactions; whether an analogue holds for IRF densities has not been checked, and the diagonal control sites make a direct transfer doubtful. If $Q_5 \equiv 0$ the hierarchy is empty beyond H, which is why the vacuity test exists.

6 Tier E — steps that are not theorems

E1. `modStd(I,0)` is correct only with high probability (no final verification). Use the verified variant for anything reported as a result.

E2. Emptiness modulo two primes is probabilistic: D7 bounds the number of bad primes but does not identify them.

E3. A small least-squares residual is not a proof that a real point exists, and a failed search is not a proof of absence.

E4. One sampled point represents its component only if it is generic in the sense of C6 and C7. Least squares from random starts does not obviously sample a density on the component; compare at least two points.

E5. The absolute thresholds 10^{-8} are scale-dependent; the rule "at least four local charges" is empirical (OUT OF SCOPE).

E6. The verdict "candidate" certifies the boost condition and, through D9, a commuting hierarchy. It certifies neither Yang–Baxter integrability nor an infinite family of non-trivial charges.

7 Open obligations (reserved for the student)

O1 (triviality)**.** A soundness theorem of the form "every component not labelled trivial is genuinely interacting" needs a definition of *trivial* first. Once it is fixed, each test needs a statement in the style of C2(i)–(ii) and C6. The existing tests detect two-locality and vacuity only literally, i.e. up to telescoping; whether that suffices is part of the question.

O2 (model equivalence)**.** For every transformation that is to be quotiented out, two statements are needed: invariance of the condition (as B7, B8 and B10 provide for telescoping, identity shifts, reflections and relabelings), and an account of which component dimensions the transformation absorbs.

8 Corrections to the reference and to earlier advice

1. **Periodic re-check.** The reference claims that $L = 7$ certifies the infinite-chain condition. It does not (C5). The development plan's habit of checking at $L = 7$ and $L = 8$ remains useful

for detecting wrap-around artefacts and bugs, but anything called new must be re-checked at $L \geq 9$, or with an independent numerical canonical form.

2. **Earlier thresholds.** The canonical-form note gave $L \geq 2n - 1 = 11$. That bound is sharp for arbitrary range-6 densities, but the residual is a commutator; the correct value for it is $L \geq 9$ (C5(c)). The value $L \geq 10$ quoted in the preceding chat message is Hokkyo's general bound and is not sharp for $(3, 4)$.

3. **Conservativeness of the classification.** The reference states that the exact classification may call a trivial piece genuine but never the converse. This holds at the level of complex components only; statements about real points need C7.

4. **Defect D1.** The `'irf'` correction must be anti-Hermitian (B4); already fixed in the rewrite, and OUT OF SCOPE in any case.

9 Suggested order of writing

C1 and C2 are done (canonical-form note). C3 should be written before the M3 checkpoint, C4 at M4, and C5 and C7 before anything is called new at M5. The Tier B lemmas are best written next to the function that uses them, as part of its acceptance test.

A Numerical checks performed for this note

All checks use dense NumPy matrices written independently of irfsolve.

item	check	result
B3	open chain, $M = 8$, $d = 2$: $[B_H, H]$ vs. $-\sum_j \sum_s s[h_j, h_{j+s}]$, general h	$1.5 \cdot 10^{-13}$
B3	same, IRF h: $[B_H, H] + Q_5$	$4.4 \cdot 10^{-14}$
B7	random IRF h, diagonal g: canonical coordinates of $Q_5(h') - Q_5(h) - [H, G]$, $d = 2$	$3.6 \cdot 10^{-15}$
B7	same, $d = 3$	$3.3 \cdot 10^{-15}$
C5(a)	random dense $(3, 4)$ densities, $d = 2$, $L = 6$ and $L = 7$	$4.3 \cdot 10^{-14}$, $2.9 \cdot 10^{-14}$
C5(b)	core-to-orbit map, ranges $n = 2, \ldots, 6$	injective exactly for $L \geq 2n - 1$

C5(c), commutators at $d = 2$. The span S of all $[\Phi, \Psi]$ with Φ, Ψ cores of length at most a and b was computed exactly in the Pauli basis; the table lists the smallest L at which wrap_L is injective on S, next to Hokkyo's sufficient bound $2(a + b) - 4$.

(a, b)	$\dim S$	smallest injective L	$2(a + b) - 4$
$(2, 2)$	32	4	4
$(2, 3)$	160	6	6
$(3, 3)$	514	7	8
$(3, 4)$	2652	9	10

For $(3, 4)$ the rank of wrap_L on S is 694, 1969, 2602, 2652 for $L = 6, 7, 8, 9$. Whether a residual of IRF form actually realises a kernel element at $L = 8$ was not tested; the statement in C5 is about what a periodic check can certify, not about whether it fails in practice.

B Index

item	tier	milestone	functions of the reference
A1–A3	A	M2	`pad`, `mul`, `ti_comm`
A4	A	M1/M3	`IRFAnsatz`, `_check_rational`, `_entry`
A5, A6	A	M3	`generate`, `linear_reduce`, `_independent_linear`
A7	A	M1/M4	`restrict`, `ab_hermitian`
A8–A10	A	M4	`_forms`, `_consistent`, `patterns`, `probe_chart`
B1, B2	B	M2	`ti_comm`, `canonical`
B3, B4	B	M3	`generate`
B5, B6	B	M3/M4	`stratum`
B7, B8	B	M3	`aux_density`, `gauge_charge_extra`
B9	B	M2/M3	`canonical`, `generate`
B10, B11	B	out of scope	`reflection`, `relabel`, `local_charges`
C1, C2	C	M2	`canonical`, `_forms`, `_canon_coeffs`
C3	C	M3	`generate`
C4	C	M4	`stratum`, `solve_chart`
C5	C	M1/M5	`verify_periodic`
C6	C	M5	`vacuous_up_to_correction`
C7	C	M5	`analyse_component`
D1–D5, D7	D	M4	Singular back end, `probe_chart`, `solve_chart`
D6, D8, D9	D	M5	`analyse_component`
E1–E6	E	M4/M5	`solve_chart`, `analyse_component`

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